IIT HYDERABAD

CS5903 – Distributed AI Training

AdaptEngine

3.Adaptive Learning, Diagnostic Assessment and Personalized Path-Planning Engine

Week 3 Report

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1 Introduction

The objective of Week 3 was to complete the adaptive diagnostic (M2), connect the mastery model to an established library through a thin adapter, and strengthen the foundation built in Week 2 so that its boundaries are safe. The finalized architecture and implementation plan were also documented inside the repository, and a single verification command was added so that every result can be reproduced and checked.

Week 3 Objective

Deliver the bounded adaptive diagnostic (M2) on top of a hardened foundation: strict input validation, mandatory content review, evidence-based mastery semantics, atomic persistence, a pinned pyBKT adapter with a parity test, and a reproducible verification run.

1.1 Status Summary

Status	Work
Completed	Adaptive diagnostic (M2) with expected-information-gain selection, caps, stopping rule, descendant skipping and cold-start preservation; mastery-model adapter with pyBKT parity; evidence-state semantics and attempt idempotency; strict validation and mandatory review gate; atomic persistence; architecture and implementation documentation; `make w3` verification runner.
Pending	Personalized planner and feasibility oracle (M3), difficulty policy (M4), review and remediation (M5), replanning (M6), decision traces (M7), simulator and comparative evaluation (M8).

Table 2: End-of-Week-3 status.

2 Response to Week 2 Feedback

The Week 2 review accepted the foundation as real progress but found problems at its boundaries and in the model semantics.

- **Review gate bypassed by the CLI.** Content review is now required by default at every live entry point, including the API and each CLI command. Loading unreviewed content needs an explicit authoring-only request. (Section **??**).
- **Unchecked event and state types.** Central validators now reject non-boolean responses such as the string `"false"`, unknown event kinds, and non-finite or out-of-range values before any state or log change.
- **Unknown prerequisites created phantom nodes.** Prerequisite references are checked before graph edges are built, and the JSON root and field types are validated, so invalid curricula fail with typed errors.
- **Lessons set `assessed` and retries were double-counted .** Instruction exposure, observed evidence and evidence-mastery are now separate. Attempts are idempotent through attempt identifiers, and practice time never moves backwards.

- **Unvalidated configuration.** The configuration is validated at load and the learning rate is bounded to $[0, 1]$, so probabilities can no longer leave the valid range.
- **Two-file persistence risk .** Learner state and log are now saved as one atomic snapshot, written to a temporary file and then replaced, and verified by a failure-injection test.
- **Model reuse (FB001).** A pinned pyBKT adapter was added behind a common mastery-model interface with a parity test. The analytical update is kept as an independent oracle.
- **Uncertainty semantics and diagnostic cap (FB002).** Entropy is now interpreted as uncertainty about a binary latent state. The diagnostic has numeric caps, a stopping rule, cold-start preservation and tested surprise-response behaviour.
- **Stale records (FB003).** The decision records and the 23-ID traceability register were synchronized with the current design, and architecture and implementation documents were added to the repository.

3 Mastery Model

Mastery is modelled with an analytical Bayesian Knowledge Tracing formulation parameterized by item difficulty $d \in [0.05, 0.95]$. For each concept, p denotes the probability that the learner has mastered it.

3.1 Guess, Slip and Predicted Success

$$g(d) = 0.30\,(1 - d), \qquad s(d) = 0.05 + 0.25\,d,$$
$$P_c(p, d) = p\,(1 - s) + (1 - p)\,g.$$

Easier items have a higher guess probability and a lower slip probability, and harder items the reverse.

3.2 Bayesian Update and Learning Transition

$$p'_{\text{correct}} = \frac{p(1-s)}{p(1-s) + (1-p)g}, \qquad p'_{\text{wrong}} = \frac{ps}{ps + (1-p)(1-g)},$$
$$p'' = p' + (1 - p')\,T, \qquad T_{\text{diag}} = 0,\ T_{\text{exercise}} = 0.05,\ T_{\text{lesson}} = 0.20.$$

Diagnostic questions are pure measurement ($T_{\text{diag}} = 0$). The value stored in learner state after an exercise is the post-transition value p''.

3.3 Reference Values

Table 3: Reference constants checked by the smoke run and the unit tests.

p	d	Response	T	Expected
0.50	0.50	Correct	0.00	0.8462
0.50	0.50	Wrong	0.00	0.1707

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Table 3 – continued from previous page

p	d	Response	T	Expected
0.50	0.50	Correct	0.05	0.8538
0.50	0.50	Wrong	0.05	0.2122
0.90	0.50	Wrong	0.00	0.6495

Parameter status. The values $p_0 = 0.20$, $\theta = 0.85$, T, $g(d)$ and $s(d)$ are synthetic heuristics and not fitted parameters.

4 Adaptive Diagnostic (M2)

The diagnostic produces a starting mastery estimate from a bounded number of questions. It is implemented as a new `diagnostic/` package that depends on the learner layer through the mastery-model interface only.

4.1 Uncertainty as Binary Entropy

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p), \qquad H(0) = H(1) = 0.$$

$H(p)$ measures uncertainty about the binary latent mastery state. It is not the variance of a parameter estimate, and it can increase after a surprising answer. For $p = 0.90$, $H = 0.469$ bits; after a wrong answer on a medium item, $p' = 0.6495$ and $H = 0.935$ bits. The model becomes more uncertain because a likely-mastered learner failed an average question. For this reason entropy is used only as a selection score and as a stop signal together with a hard item cap.

4.2 Expected Information Gain

For the active concept, the diagnostic evaluates every exercise not yet asked in the session and selects the one with the largest expected reduction in entropy. Using g, s, P_c and the posteriors p'_{correct}, p'_{wrong} from Section 3, the score for a candidate exercise a is

$$\text{EIG}(a) = H(p) - \Big[P_c \, H\big(p'_{\text{correct}}\big) + (1 - P_c) \, H\big(p'_{\text{wrong}}\big) \Big], \tag{1}$$

where $H(p)$ is the current binary entropy and the bracketed term is the expected entropy after observing the response. The selected item is

$$a^* = \arg \max_{a \in \mathcal{A}_{\text{unasked}}} \text{EIG}(a).$$

Operational details.

- The score is computed without the learning transition ($T_{\text{diag}} = 0$), since the diagnostic measures existing mastery and does not model learning during the test.
- Ties are broken by the difficulty closest to 0.5, then by lexicographic `activity_id`, so selection is deterministic.
- Under the current guess and slip functions, EIG tends to be largest for the hardest items, because a correct answer to a hard item is strong evidence of mastery when guessing is rare. This is a property of the heuristic parameters and is noted as a limitation.

4.3 Rules and Bounds

Table 4: Diagnostic rules.

Rule	Definition
Concept order	Stable topological order, ties broken by concept identifier.
Global cap	At most $N_{\max} = 8$ items in total.
Concept cap	At most 3 items per concept.
Entropy stop	A concept stops when $H(p) \leq 0.50$, i.e. $p \leq 0.11$ or $p \geq 0.89$.
Descendant skipping	If a concept ends with $p < 0.40$, its descendants are not probed and remain `UNASSESSED`.
Cold start	Unprobed concepts keep the prior $p_0 = 0.20$, are marked `UNASSESSED`, are treated as unmastered, and existing learner state is never overwritten.
Tie-break	Equal EIG: smaller $\lvert d - 0.5 \rvert$, then lexicographic `activity_id`.
Stop reasons	`ENTROPY`, `CONCEPT_CAP`, `GLOBAL_CAP`, `NO_ITEMS`, `SKIPPED_ANCESTOR_UNMASTERED`.

5 Mastery Model Adapter

5.1 Interface

The `MasteryModel` protocol defines `predict_correct(p, d)` and `update(p, d, correct, transition)`. Two implementations exist:

- `CustomBKT`**:** the analytical equations of Section 3, kept as the independent oracle.
- `PyBKTAdapter`**:** wraps the pinned pyBKT library.

5.2 Difficulty Classes

pyBKT works with discrete item classes, while $g(d)$ and $s(d)$ are continuous. Difficulty is therefore mapped explicitly to three classes, each using its representative difficulty.

Class	Range	d_{rep}	g	s
Easy	$d < 0.35$	0.20	0.24	0.100
Medium	$0.35 \leq d < 0.65$	0.50	0.15	0.175
Hard	$d \geq 0.65$	0.80	0.06	0.250

Table 5: Mapping from continuous difficulty to pyBKT item classes.

5.3 Parity and Isolation

At the representative difficulties 0.20, 0.50 and 0.80, the adapter matches the analytical oracle within 10^{-9}. Parameters are never swapped silently, and the mapped values are reported above.

pyBKT is imported only in `learner/adapter.py`, and an import-scanning test fails if it appears anywhere else.

6 Evidence-State Semantics

Previously, completing a lesson set the concept as assessed, so replaying one lesson eight times raised p above θ without a single question answered. Learner state now separates instruction exposure from observed evidence.

Table 6: Learner-concept state (immutable).

Field	Meaning
`p`	Mastery belief in $[0, 1]$.
`lessons_done`	Exposure count; lessons shift the prior with $T_{\text{lesson}} = 0.20$.
`responses`	Number of observed exercise and diagnostic attempts.
`correct_responses`	Correct attempts, at most `responses`.
`status`	`UNASSESSED` until at least one exercise or diagnostic response.
`seen_attempt_ids`	Attempt identifiers already applied.
`half_life_days`, `last_practice_day`	Review-related state.

- **Lessons** never increase `responses` and never set `ASSESSED`.
- **Evidence-mastered:** a concept counts as mastered only when $p \geq \theta$ and `responses` ≥ 2. Prerequisite gating uses this definition.
- **Idempotent attempts:** an event whose `attempt_id` was already applied is logged as `DUPLICATE_IGNORED` and leaves the state unchanged.
- **Monotonic time:** `last_practice_day` is the maximum of the old value and the event day, so a late-arriving earlier event cannot rewind history.

7 Architecture and Documentation

7.1 Layering

Dependencies run in one direction, and a test enforces this rule.

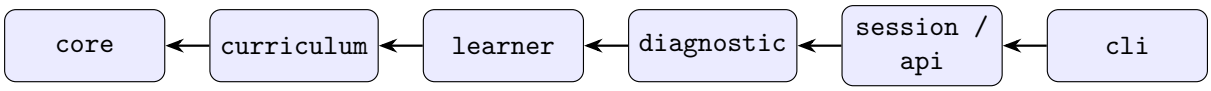


Figure 1: Module layering

Third-party libraries are isolated: NetworkX is imported only in `curriculum/graph.py` and pyBKT only in `learner/adapter.py`.

7.2 Documentation Added

Table 7: Documentation and records now kept in the repository.

File	Content
`docs/ARCHITECTURE.md`	Implemented and planned modules, layering, design principles, components, workflow, data model, optional agentic layer.
`docs/MASTERY_MODEL.md`	Equations, reference values, parameter status, entropy interpretation, evidence-state rules.
`docs/DIAGNOSTIC.md`	M2 specification, bounds and test guarantees.
`docs/IMPLEMENTATION_PLAN.md`	Weekly plan, risks and fallbacks.
`docs/TECHNICAL_FEASIBILITY.md`	Tools, datasets, assumptions and limitations.
`docs/reuse_inventory.md`	Library versions, licences and mapping notes.
`records/TRACEABILITY.md`	Authoritative register of 23 test IDs (TC1.1 to TC8.2).

8 Test Register and Milestone Status

8.1 Test Cases Targeted in Week 3

Table 8: Week 3 acceptance tests.

ID	Scenario and expected result
TC1.2	Valid and invalid curriculum, state, event and configuration fixtures, and unreviewed content: valid ones load, invalid ones are rejected with typed errors before any change to state or log.
TC2.1	Diagnostic stops at the entropy threshold or item cap and never exceeds $N_{\max}$.
TC2.2	Concepts without previous observations stay `UNASSESSED` with prior state preserved.
TC4.1	Mastery references 0.1707 and 0.2122; adapter matches the oracle.

8.2 Milestone Progress

Table 9: Milestone status at the end of Week 3.

Milestone	Status	Note
M1	Done	Boundaries validated and review enforced; TC1.1 and TC1.2 cover loading and rejection.
M2	Implemented	Diagnostic with caps, stopping, skipping and cold start; TC2.3 depends on the planner and moves to Week 4.

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Milestone	Status	Note
M3	Pending	Planner, feasibility bounds and oracle in Week 4.
M4	Partial	Mastery update and adapter done; next-item difficulty policy in Week 4.
M5	Pending	Week 5.
M6	Pending	Week 5.
M7	Pending	Weeks 5–6.
M8	Pending	Weeks 6–7.

9 Design Choices

The following values were fixed so that work could proceed. They are recorded in `records/DECISIONS.md`.

ID	Decision	Value applied
D1	Global diagnostic item cap	8
D2	Per-concept item cap	3
D3	Entropy stop threshold	$H(p) \leq 0.50$
D4	Responses required for evidence-mastery	2
D5	Stored value after an exercise	post-transition p''
D6	pyBKT difficulty classes	3 (easy, medium, hard)
D7	Effect of lessons	shift prior only; never assessed
D8	Descendant skipping	ancestor $p < 0.40$
D10	Tie-break	smaller $\lvert d - 0.5 \rvert$, then `activity_id`

Table 11: Decisions applied in Week 3.

10 Limitations and Risks

- **Heuristic parameters.** All model constants are synthetic and unfitted. A sensitivity run is planned for the evaluation stage.
- **Class-based adapter.** pyBKT uses three discrete item classes, so the adapter matches the continuous model exactly only at the representative difficulties.
- **Scripted personas.** The diagnostic demonstration uses hand-written learners. Estimation quality against fixed and random selection has not been measured yet and is planned for the M8 harness.
- **Binary responses.** Partial credit, hints and multi-concept items are outside the prototype.
- **Review quality.** A hash check shows that content was not altered after review, but does not establish the quality of the review.

11 Plan for Week 4

1. Build the tiny static plan and then the eligibility filter with deterministic ranking driven by the diagnostic output.
2. Implement day-by-day planning for fixed time budgets.
3. Add the next-item difficulty policy based on predicted success probability.
4. Add feasibility bounds, the exact oracle and the `INFEASIBLE` and `UNKNOWN` outcomes.
5. Extend the curriculum with a larger fixture.
6. Target tests: TC2.3, TC3.1–TC3.6 and TC4.2.